

Incident Documentation

Case: Ransomware Incident Investigation

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1. Executive Summary

On **August 24, 2016**, Bob Smith’s workstation (**we8105desk**) was infected by **Cerber ransomware** after a USB key named **MIRANDA_PRI** was used. The infection began when the machine visited the domain **solidaritedeproximite.org** to download a file named **/mhtr.jpg**, which used **steganography** to hide malicious code inside an image. A VB script (length **4490**) was executed via **wscript.exe** to launch **121214.tmp** (PPID: **3964**), starting the encryption process. The malware encrypted **406 text files** within the user profile and extended its reach to a network file server at **192.168.250.20**, where it encrypted **257 distinct PDFs**. Finally, the ransomware directed the victim to the domain **cerberhhyed5frqa.xmfir0.win** to facilitate the ransom payment.

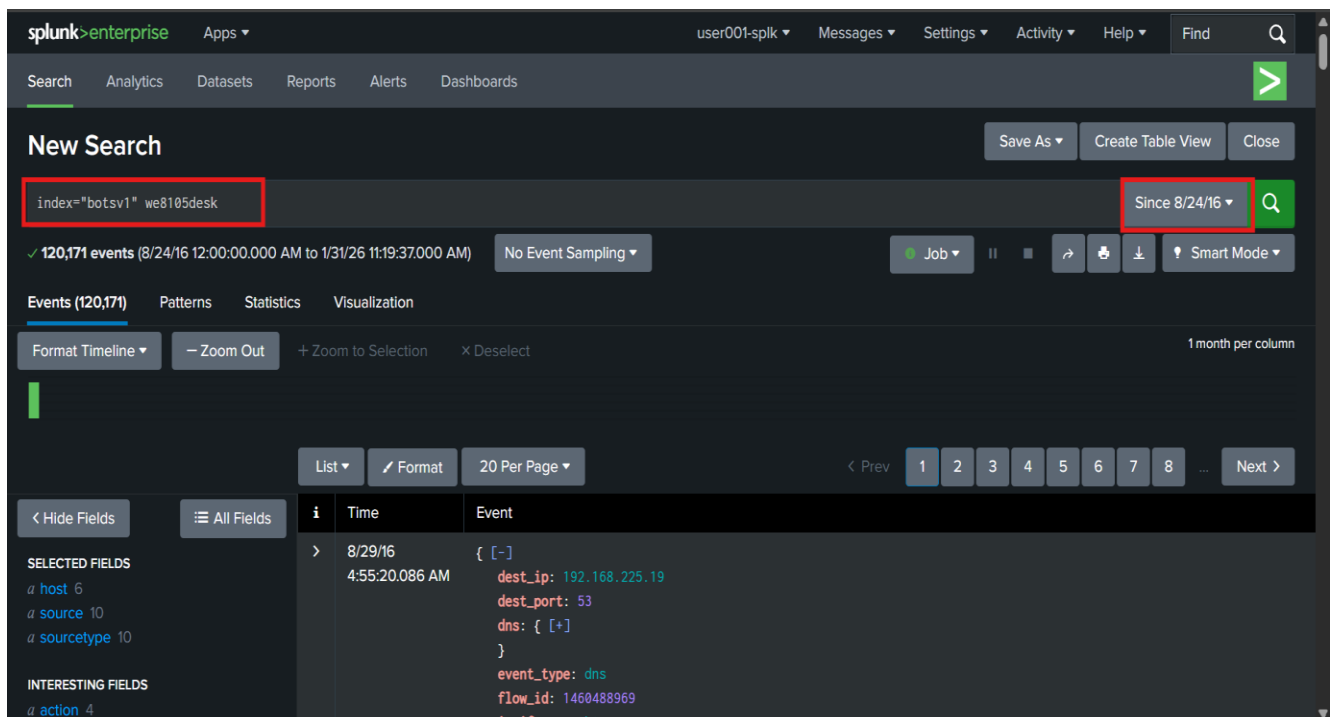
2. Tools

Tool	Purpose
Splunk	Centralized platform for collecting, searching, and analyzing security logs.
Screenshots	Evidence capture for visual documentation
Sysmon (Windows)	Detailed logging of process, network, and system activity on Windows hosts
Splunk Stream	Network traffic visibility through real-time flow and protocol metadata.
Windows Registry	Persistent storage of system and user configuration artifacts.
Suricata	Network-based intrusion detection and threat alerting engine.

3. Questions & Answers

1. What was the most likely IPv4 address of we8105desk on 24AUG2016?

By restricting the Splunk time range to August 24, 2016, I identified the victim's IP address by searching the **botsv1** index for the workstation **we8105desk**. By analyzing the **src_ip** field within the network logs, I confirmed that the IP address **192.168.250.100** was assigned to the machine. This was the source of the high-volume traffic communicating with the malicious domains and the internal file server during the ransomware outbreak.



*Splunk search interface showing the query **index="botsv1" we8105desk** to isolate all events associated with the infected workstation.*

The screenshot shows the Splunk Search interface. On the left, the 'Fields' list is visible, with 'src_ip' selected. The main panel displays the 'Reports' tab for the 'src_ip' field. The table shows the top 10 values, with the first row, '192.168.250.100', highlighted in red. The table has columns for the value, count, and percentage. The right sidebar shows the search criteria: 'a.json sourcetype = suricata'.

Top 10 Values	Count	%
192.168.250.100	52,361	88.042%
192.168.2.50	2,744	4.614%
127.0.0.1	66	0.111%
192.168.250.255	62	0.104%
0.0.0.0	40	0.067%
192.168.225.111	35	0.059%
192.168.229.203	21	0.035%
192.168.225.96	17	0.028%
192.168.224.97	12	0.02%
192.168.224.88	11	0.018%

*Filtered Splunk results highlighting the **src_ip** field and the identified victim IP address **192.168.250.100***

2. Amongst the Suricata signatures that detected the Cerber malware, which one alerted the fewest number of times? Submit ONLY the signature ID value as the answer.

I searched the **botsv1** index on **August 24, 2016**, using the victim's IP address and the keyword **"cerber"**. This search returned **exactly three events**. I manually investigated each of these three events one by one, and by checking the last one, I found **the signature ID 2816763**, which identified the alert that triggered the fewest number of times.

The screenshot shows the Splunk Enterprise interface. At the top, the search bar contains the query: `index="botsv1" sourcetype="suricata" src_ip=192.168.250.100 "cerber"`. The search results show 3 events. The first event is expanded, showing the following details:

Time	Event
8/24/16 5:15:12.668 PM	<pre>{ [-] alert: { [+] } dest_ip: 192.168.250.20 dest_port: 53 event_type: alert flow_id: 4097325782 to: 192.168.250.100 }</pre>

The interface also shows a list of fields on the left, including `host`, `source`, `sourcetype`, and `action`.

*Splunk search query using **index=botsv1 sourcetype=suricata "cerber"** to identify the specific threat signatures and filter for the least frequent alert.*

< Hide Fields		≡ All Fields	List ▾	Format	20 Per Page ▾
a vendor 1					
# vendor_gid 1					
# vendor_rev 2					
# vendor_sid 2					
+ Extract New Fields					
i	Time	Event			
		Event	action ▾	allowed	▾
			alert.action ▾	allowed	▾
			alert.category ▾	A Network Trojan was detected	▾
			alert.gid ▾	1	▾
			alert.rev ▾	4	▾
			alert.severity ▾	1	▾
			alert.signature ▾	ETPRO TROJAN Ransomware/Cerber Checkin 2	▾
			alert.signature_id ▾	2816763	▾
			alert_gid ▾	1	▾
			alert_rev ▾	4	▾
			category ▾	A Network Trojan was detected	▾
			dest_ip ▾	85.93.0.0	▾
			dest_port ▾	6892	▾
			dvc ▾	suricata-ids.waynecorpinc.local	▾
			event_type ▾	alert	▾
			eventtype ▾	suricata_eve_ids_attack (attack_ids)	▾
			flow_id ▾	818473913	▾
			ids_type ▾	network	▾
			in_iface ▾	eth1	▾

Splunk event details displaying **Signature ID 2816763** as the identified alert.

3. What fully qualified domain name (FQDN) does the Cerber ransomware attempt to direct the user to at the end of its encryption phase?

A closer examination of DNS traffic revealed indicators associated with **Cerber ransomware** when filtering on the victim's IP address and the keyword **cerber*** knowing that **Cerber** uses **DNS** to **direct victims** to its **payment site**, I filtered for the **stream:dns sourcetype** and excluded common noise such as **Microsoft** and **Google domains**. After manually reviewing the remaining event, I identified the fully qualified domain name (FQDN) as **cerberhhyed5frqa.xmfir0.win**, which the ransomware directs users to at the end of the encryption phase.

splunk>enterprise Apps ▾ user001-splk ▾ Messages ▾ Settings ▾ Activity ▾ Help ▾ Find 🔍

Search Analytics Datasets Reports Alerts Dashboards >

New Search

Save As ▾ Create Table View Close

index="botsv1" sourcetype="stream:dns" src_ip=192.168.250.100 "cerber*" Since 8/24/16 🔍

✓ 1 event (8/24/16 12:00:00.000 AM to 1/31/26 11:46:19.000 AM) No Event Sampling ▾ Job ▾ || ▢ ▸ ⬵ ⬇ ⬆ Smart Mode ▾

Events (1) Patterns Statistics Visualization

Format Timeline ▾ - Zoom Out + Zoom to Selection x Deselect 1 month per column

List ▾ Format 20 Per Page ▾

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SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
a app 1

i	Time	Event
>	8/24/16 5:15:12.668 PM	{ [-] bytes: 90 bytes_in: 90 bytes_out: 0 dest_ip: 192.168.250.20 dest_mac: 08:0C:29:C3:C4:00 dest_port: 53 edtime: 2016-08-24T17:15:12.668000Z

*Splunk search results for **sourcetype="stream:dns"** used to identify the specific fully qualified domain name (FQDN) assigned for the ransom payment.*

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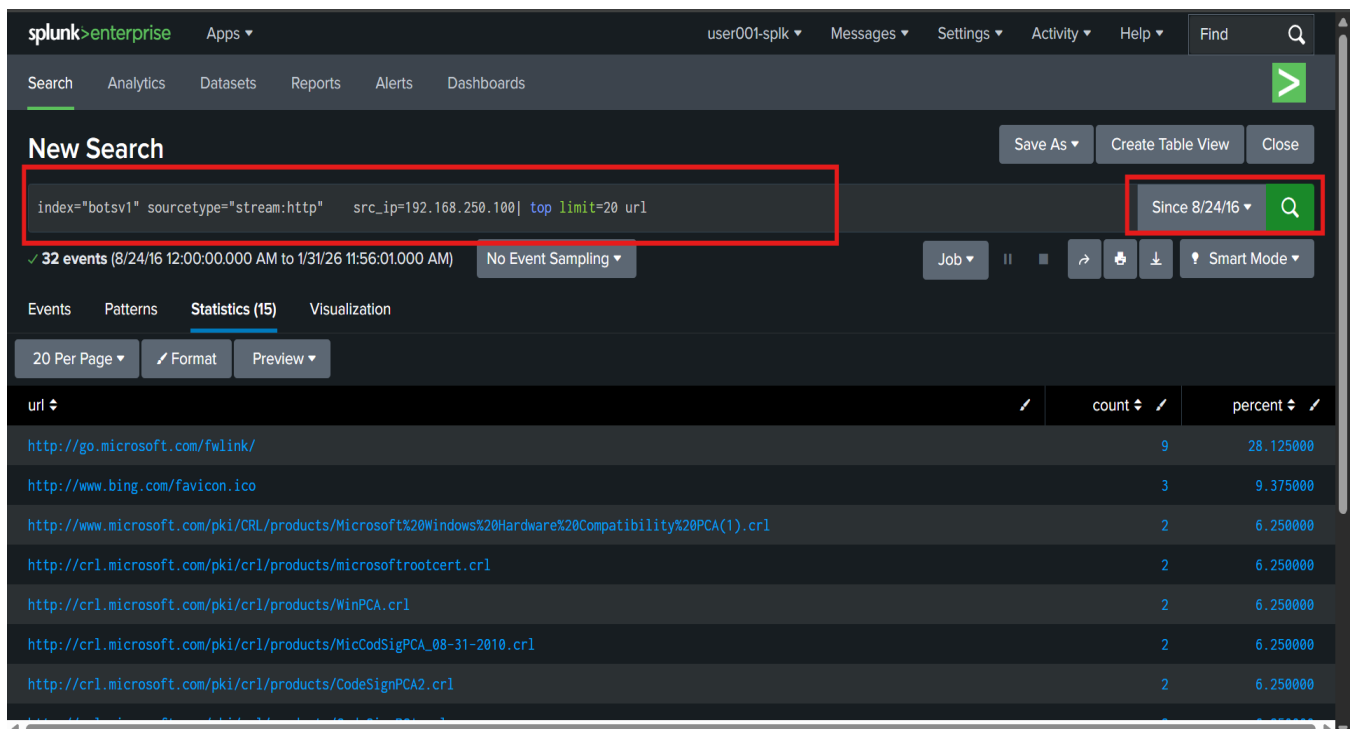
timeendpos 1
a timestamp 1
timestamppos 1
transaction_id 1
a transport 1
+ Extract New Fields

i	Time	Event
		message_type ▾ QUERY ▾
		message_types ▾ QUERY ▾
		message_type[] ▾ QUERY ▾
		protocol ▾ dns ▾
		queries ▾ cerberhhyed5frqa.xmfir0.win ▾
		query ▾ cerberhhyed5frqa.xmfir0.win ▾
		query_type[] ▾ A ▾
		query[] ▾ cerberhhyed5frqa.xmfir0.win ▾
		record_type ▾ A ▾
		src ▾ 192.168.250.100 ▾
		src_ip ▾ 192.168.250.100 ▾

*Splunk event details displaying the unique query field containing the FQDN **cerberhhyed5frqa.xmfir0.win**.*

4. What was the first suspicious domain visited by we8105desk on 24AUG2016?

The investigation began with a deep dive into **the botsv1 index**, focusing on the **stream:http sourcetype**. By isolating **the url field**, I was able to reconstruct the sequence of **web requests** generated by the workstation. Analyzing **the earliest entries** revealed **solidaritedeproximityte.org** as the first suspicious domain accessed by **we8105desk**, marking the initial point of the infection chain.



The screenshot shows the Splunk Enterprise interface. The search bar contains the query: `index="botsv1" sourcetype="stream:http" src_ip=192.168.250.100 | top limit=20 url`. The search results show 32 events. The table below displays the top results for the `url` field.

url	count	percent
http://go.microsoft.com/fwlink/	9	28.125000
http://www.bing.com/favicon.ico	3	9.375000
http://www.microsoft.com/pki/CRL/products/Microsoft%20Windows%20Hardware%20Compatibility%20PCA(1).crl	2	6.250000
http://crl.microsoft.com/pki/crl/products/microsoftrootcert.crl	2	6.250000
http://crl.microsoft.com/pki/crl/products/WinPCA.crl	2	6.250000
http://crl.microsoft.com/pki/crl/products/MicCodSigPCA_08-31-2010.crl	2	6.250000
http://crl.microsoft.com/pki/crl/products/CodeSignPCA2.crl	2	6.250000

*Splunk search query using **index=botsv1 sourcetype="stream:http"** we8105desk to analyze the sequence of URLs and identify the initial suspicious domain.*

20 Per Page ▾			Format	Preview ▾
url ↕		count ↕		percent ↕
http://go.microsoft.com/fwlink/		9		28.125000
http://www.bing.com/favicon.ico		3		9.375000
http://www.microsoft.com/pki/CRL/products/Microsoft%20Windows%20Hardware%20Compatibility%20PCA(1).crl		2		6.250000
http://crl.microsoft.com/pki/crl/products/microsoftrootcert.crl		2		6.250000
http://crl.microsoft.com/pki/crl/products/WinPCA.crl		2		6.250000
http://crl.microsoft.com/pki/crl/products/MicCodSigPCA_08-31-2010.crl		2		6.250000
http://crl.microsoft.com/pki/crl/products/CodeSignPCA2.crl		2		6.250000
http://crl.microsoft.com/pki/crl/products/CodeSignPCA.crl		2		6.250000
http://crl.microsoft.com/pki/crl/products/CodeSigPCA.crl		2		6.250000
http://solidaritedeproximity.org/mhtr.jpg		1		3.125000
http://shell.windows.com/fileassoc/fileassoc.asp		1		3.125000
http://shell.windows.com/HeaderSlice.jpg		1		3.125000
http://shell.windows.com/0409/fileassoc.css		1		3.125000
http://ipinfo.io/json		1		3.125000
http://92.222.104.182/mhtr.jpg		1		3.125000

*Splunk event details highlighting **solidaritedeproximity.org** as the initial domain identified in the HTTP traffic sequence.*

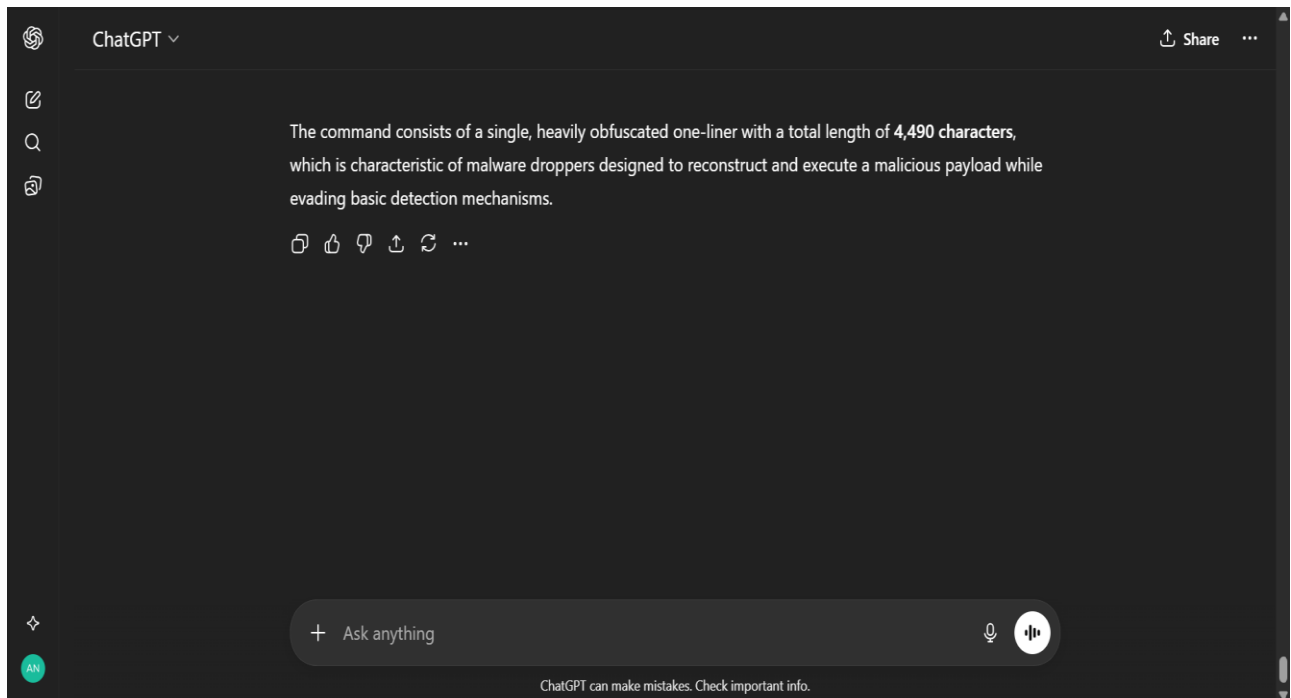
5. During the initial Cerber infection a VB script is run. The entire script from this execution, pre-pended by the name of the launching .exe, can be found in a field in Splunk. What is the length of the value of this field?

The focus of the analysis then changed to **determine the length of the VBScript code** utilized for the attack by delving into **the botsv1 index** and looking at the **Sysmon logs**, specifically **(Event ID 1) = Process Creation**. I focused my analysis on the **command-line** usage of the scripts, particularly if **wscript.exe** or **cscript.exe** is immediately used after executing **the malicious document**. By examining the **CommandLine** field, which **contained the entire script** and had the name of the executable at the beginning, I was able to identify the number of characters in the script, which had **4490 characters**.

The screenshot shows the Splunk search interface. The search bar contains the query: `index="botsv1" sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" we8105desk ("wscript.exe" OR "cscript.exe" OR "cmd.exe") CommandLine`. The search results show 46 events. The first event is displayed in a table view, showing the event details for Event ID 1 (Process Creation) on 8/24/16 at 5:15:29.000 PM. The event data includes the process name, GUID, and command line arguments.

Time	Event
8/24/16 5:15:29.000 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385F-C22A-43E0-BF4C-06F5698FFB09}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2016-08-24T17:15:29.335474700Z' /><EventRecordID>423923</EventRecordID><Correlation><Execution ProcessID='1216' ThreadID='1768' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>we8105desk.waynecorpinc.local</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='UtcTime'>2016-08-24 17:15:29.325</Data><Data Name='ProcessGuid'>{0F2D76F0-D631-578D-0000-0010F14B9700}</Data><Data Name='ProcessId'>4140</Data><Data Name='Image'>C:\Windows\System32\IP

*Splunk search query targeting **Sysmon Event ID 1** and **script execution engines** to extract the command-line arguments.*



ChatGPT's analysis of the script, confirming a total length of 4490 characters.

6. What is the name of the USB key inserted by Bob Smith?

To determine whether **removable media** had been connected to the system, **registry activity** associated with **we8105desk** was analyzed. Using the **WinRegistry sourcetype** in the **botsv1** index, I searched for the **friendlyname** registry key, which stores **human-readable names** for **connected devices**. This analysis revealed the **USB** device name as **MIRANDA_PRI**.

The screenshot shows the Splunk Enterprise search interface. The search bar contains the query: `index="botsv1" sourcetype="winregistry" "friendlyname"`. The search results show 2 events. The first event is expanded, showing the following details:

Time	Event
8/24/16 4:42:17.000 PM	<pre>08/24/2016 10:42:17.287 ... 2 lines omitted ... process_image="c:\Windows\System32\svchost.exe" registry_type="SetValue" key_path="HKLM\system\controlset001\enum\wpdusenumroot\umb\2&37c186b&0&storage#volume#_??_usbstor#disk&ven_generic &prod_flash_disk&rev.8.07#7d961196&0#\friendlyname" data_type="REG_SZ"</pre>

*Splunk search query targeting the **WinRegistry sourcetype** to locate the **FriendlyName** key and identify connected hardware.*

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All Fields

List

Format

20 Per Page

INTERESTING FIELDS

a action 1

a data 1

a data_type 1

a dest 1

a event_status 1

a eventtype 1

a index 1

a key_path 2

linecount 1

a msg 1

a object 1

a object_category 1

a object_path 2

pid 2

a process_image 2

a punct 2

a registry_key_name 2

a registry_path 2

a registry_type 1

a registry_value_data 1

a registry_value_name 1

a registry_value_type 1

a splunk_server 1

a status 1

a tag 4

i

Time

Event

data_type="REG_SZ"

Show all 8 lines

Event Actions

Type	Field	Value	Actions
Selected	host	we8105desk	
	source	WinRegistry	
	sourcetype	WinRegistry	
Event	action	modified	
	data	MIRANDA_PRI	
	data_type	REG_SZ	
	dest	we8105desk	
	event_status	(0)The operation completed successfully.	
	eventtype	winregistry_windows (change endpoint os windows)	
	key_path	HKLM\system\controlset001\enum\wpdbusenumroot\umb\2&37c186b&0&storage#volume#_??_usbstor#disk&ven_generic&prod_flash_disk&rev_8.07#7d961196&0#friendlyname	
	msg	The operation completed successfully.	
	object	friendlyname	
	object_category	registry	

Splunk event details displaying the **registry_value_name** as **FriendlyName** with the actual device name **MIRANDA_PRI**.

7. Bob Smith's workstation (we8105desk) was connected to a file server during the ransomware outbreak. What is the IPv4 address of the file server?

The analysis then shifted to **network file-sharing activity** by examining the **stream:smb** sourcetype in the **botsv1** index, as **SMB traffic** commonly indicates access to shared network resources. Focusing on **outbound connections** from the infected workstation **we8105desk** over **port 445**, I reviewed the **destination IP addresses** involved in the communication. This analysis identified **192.168.250.20** as the **file server**, confirmed as the **target system** with which the workstation interacted **while encrypting files** across the network during the outbreak.

The screenshot displays the Splunk Enterprise search interface. The search bar contains the query: `index="botsv1" sourcetype="stream:smb" src_ip=192.168.250.100 dest_port = 445`. The search results show 38,815 events from 8/24/16 12:00:00.000 AM to 1/31/26 12:26:38.000 PM. The event list shows a single event at 8/24/16 6:27:38.160 PM with the following details:

i	Time	Event
>	8/24/16 6:27:38.160 PM	<pre>{ [-] bytes: 144 bytes_in: 72 bytes_out: 72 command: [[+]] dest_ip: 192.168.250.20 dest_port: 445 }</pre>

*Splunk search query targeting **stream:smb** traffic from the infected workstation to pinpoint communication with the network file server.*

Splunk field sidebar highlighting **dest_ip** with the target file server address **192.168.250.20**.

The screenshot shows the Splunk field sidebar on the left, with the 'dest_ip' field highlighted. A pop-up window titled 'dest_ip' is open, showing the following details:

- 1 Value, 100% of events
- Selected: Yes No
- Reports: Top values, Top values by time, Rare values
- Events with this field
- Values table:

Values	Count	%
192.168.250.20	38,815	100%

The pop-up window also displays the event details for the selected field, including the following fields:

- nt_status: [(+)]
- src_ip: 192.168.250.100
- src_mac: 0C:4D:E9:AA:B1:27
- src_port: 49390
- dest_ip: 192.168.250.20
- dest_mac: 00:0C:29:C3:C4:00
- dest_port: 445
- endtime: 2016-08-24T18:27:24.795179Z
- nt_status: [(+)]
- src_ip: 192.168.250.100
- src_mac: 0C:4D:E9:AA:B1:27

8. How many distinct PDFs did the ransomware encrypt on the remote file server?

The analysis then focused on quantifying the impact of the ransomware on the remote file server by examining activity associated with **we3269srv** (192.168.250.20) in the **botsv1** index. I filtered events related to the **.pdf** file extension and narrowed the results to ransomware specific behavior by identifying file modification and rename activity. Using the (**Relative_Target_Name**) field to calculate a distinct count of affected files, I determined that **257 unique PDF files** were encrypted on the server.

The screenshot displays the Splunk Enterprise web interface. At the top, the navigation bar includes 'splunk>enterprise', 'Apps', and user information. Below this, a 'New Search' panel contains a search query: `index=botsv1 sourcetype="*" we9041srv 192.168.250.100 Relative_Target_Name="*.pdf" | dedup Relative_Target_Name | stats count`. The search results show 257 events. The 'Statistics (1)' tab is active, displaying a table with one row: 'count' with a value of 257. The interface also shows various controls like 'Save As', 'Create Table View', 'Close', and 'Smart Mode'.

count
257

*Splunk statistics table displaying the distinct count of 257 unique files identified within the **Relative_Target_Name** field.*

9. The VBscript found in question 204 launches 121214.tmp. What is the ParentProcessId of this initial launch?

To track the source of the script execution activity for the **we8105desk**, Sysmon logs for the **botsv1** index were examined. I have used the Windows binaries **wscript.exe** and **cscript.exe** to execute the script and the temporary file name **121214.tmp** to monitor the activity. The results of this query returned a **single event**, and based on the information provided within the results, the **ParentProcessId**, or **PPID**, was **3968**.

The screenshot shows the Splunk Enterprise interface with a search query: `index=botsv1 sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" we8105desk (wscript.exe OR cscript.exe) 121214.tmp`. The search results show 1 event. The event details are as follows:

Time	Event
8/24/16 4:48:21.000 PM	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385F-C22A-43E0-BF4C-06F5698FFBD9}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2016-08-24T16:48:21.552726900Z' /><EventRecordID>364394</EventRecordID><Correlation><Execution ProcessID='1216' ThreadID='1768' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>we8105desk.waynecorpinc.local</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='UtcTime'>2016-08-24 16:48:21.537</Data><Data Name='ProcessGuid'>{0F2D76F0-CFD5-57BD-0000-0010E3AC3400}</Data><Data Name='ProcessId'>1476</Data><Data Name='Image'>C:\Windows\SysWOW64\cmd.exe</Data><Data Name='Command Line'>C:\Windows\SysWOW64\cmd.exe /C START /B "C:\Users\bob\mk\we8105desk\121214.tmp"</pre>

Splunk search query targeting Sysmon Event ID 1 for the temporary file and script engine to isolate the process creation event.

< Hide Fields All Fields List Format 20 Per Page			
#	ProcessId	i	Time
a	punct		
#	RecordID		
a	SecurityID		
a	session_id		
a	SHA1		
a	SHA256		
a	signature		
#	signature_id		
a	splunk_server		
a	tag		
a	tag:eventtype		
#	Task		
#	TerminalSessionId		
a	TimeCreated		
a	user		
a	User		
a	UtcTime		
a	vendor_product		
#	Version		
+ Extract New Fields			
Event			
	LogonGuid		[0F2D76F0-C612-57BD-0000-002015F80600]
	LogonId		0x6f815
	MD5		AD7B9C14083B52BC532FBA5948342B98
	Opcode		0
	ParentCommandLine		"C:\Windows\System32\WScript.exe" "C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\20429.vbs"
	ParentImage		C:\Windows\SysWOW64\wscript.exe
	ParentProcessGuid		[0F2D76F0-CEA9-57BD-0000-0010C8133100]
	ParentProcessId		3968
	ProcessGuid		[0F2D76F0-CFD5-57BD-0000-0010E3AC3400]
	ProcessId		1476
	RecordID		364394
	SHA1		EE8CBF12D87C4D388F09B4F69BED2E91682920B5
	SHA256		17F746D82695FA9B35493B41859D39D786D32B23A9D2E00F401DEC7A02402AE
	SecurityID		S-I-5-18
	Task		1
	TerminalSessionId		1
	TimeCreated		2016-08-24T16:48:21.552726900Z

Splunk event details highlighting the **ParentProcessId** field with the specific value of **3968**.

10. The Cerber ransomware encrypts files located in Bob Smith's Windows profile. How many .txt files does it encrypt?

To assess the scope of **user data impacted**, **encrypted text files** on the workstation **we8105desk** were examined. An initial review of the **botsv1 index** returned **410 text files** in total. To isolate only the **files encrypted by Cerber** within **Bob Smith's Windows profile**, I examined **the file paths** in detail. This revealed that **4 files** were located within the **Sysmon directory**, indicating **system related artifacts** rather than **user data**. After excluding these entries, I confirmed that **406 .txt files** were encrypted within **Bob Smith's profile**.

The screenshot shows the Splunk interface with a search query: `index=botsv1 sourcetype="*" we8105desk *.txt`. The search results show 410 events. The table below lists the file paths and their counts.

TargetFilename	count
C:\Sysmon\._Eula.txt	1
C:\Sysmon\AuditPol_AFTER_WE8105DESK.txt	1
C:\Sysmon\AuditPol_BEFORE_WE8105DESK.txt	1
C:\Sysmon\Eula.txt	1
C:\Users\bob.smith.WAYNECORPINC\Desktop\2010\Office 2010 Pro\Key.txt	1
C:\Users\bob.smith.WAYNECORPINC\Desktop\2010\Project 2010\Key.txt	1

Splunk statistics showing the total count of 410 text files and the 4 specific entries identified within the system directory.

11. The malware downloads a file that contains the Cerber ransomware cryptor code. What is the name of that file?

I went back to my analysis of the first suspicious domain the machine visited, **solidaritedeproximate.org**. I remembered seeing a specific file associated with this domain during my earlier search of the **stream:http** logs. By reviewing those web requests again, I confirmed that the file being downloaded was **/mhtr.jpg**. Although it has a **JPEG extension**, this was the file that actually contained the malicious code used to initiate the encryption process.

The screenshot shows the Splunk Enterprise web interface. At the top, the navigation bar includes 'splunk>enterprise', 'Apps', and user information 'user001-splk'. Below this is a search bar with the query 'index=botsv1 sourcetype="stream:http" src_ip=192.168.250.100 solidaritedeproximate.org' and a time filter 'Since 8/24/16'. The search results show 1 event for the time range '8/24/16 12:00:00.000 AM to 1/31/26 1:34:30.000 PM'. The event details are displayed in a table with columns 'Time' and 'Event'.

Time	Event
8/24/16 4:48:13.285 PM	{ [-] accept: */* accept_language: en-us ack_packets_in: 7 ack_packets_out: 4 bytes: 2729 bytes_in: 356 }

*Splunk search query targeting **stream:http** with the filter "**solidaritedeproximate.org**" to isolate the malicious file download.*

< Hide Fields ≡ All Fields List ▾ Format 20 Per Page ▾		
i	Time	Event
		<pre> http_user_agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/4.0; SLCC2; .NET CLR 2.0.50727; 3.0.30729; Media Center PC 6.0; InfoPath.3) missing_packets_in: 0 missing_packets_out: 0 network_interface: eth1 packets_in: 10 packets_out: 8 reply_time: 480720 request: GET /mhtr.jpg HTTP/1.1 request_ack_time: 4999 request_time: 0 response_ack_time: 206654 response_time: 208407 sc_cache_control: no-store, no-cache, must-revalidate sc_date: Wed, 24 Aug 2016 16:48:12 GMT sc_pragma: no-cache server: Apache server_rtt: 4999 server_rtt_packets: 1 server_rtt_sum: 4999 site: solidaritedeproximate.org src_headers: GET /mhtr.jpg HTTP/1.1 Accept: */* Accept-Language: en-us Range: bytes=9673- </pre>

*Splunk event data showing the GET request for **/mhtr.jpg** within the **HTTP stream logs**.*

12. Now that you know the name of the ransomware's encryptor file, what obfuscation technique does it likely use?

The technique used to hide data like code, text, or files within another file (such as the **/mhtr.jpg** image used here) is **steganography**.

4. Conclusion

The forensic investigation into the infection of **we8105desk** provided a **comprehensive reconstruction** of a **ransomware outbreak** within the **botsv1 environment**. By leveraging **Splunk** and **Sysmon logs**, I was able to:

- **Map the Infection Lifecycle:** Trace the initial compromise from the suspicious download of **mhtr.jpg** from the domain **solidaritedeproximitye.org**, through the execution of a malicious script obfuscated as a temporary file (**121214.tmp**).
- **Identify Lateral Impact:** Analyze **SMB traffic** to pinpoint **the encryption of 257 unique PDF files** on the remote file server (**192.168.250.20**) and the compromise of **406 user specific text files** within the workstation's local profile.
- **Expose Attacker Artifacts:** Uncover technical indicators such as the **ParentProcessId (3968)**, the use of **wscript.exe** for script execution, and the connection of removable media identified as **MIRANDA_PRI**.
- **Establish Attribution and Intent:** Confirm the presence of **Cerber ransomware through file modification patterns** and the specific character counts calculated within **malicious script blocks**.

This investigation demonstrates a **structured, evidence based threat hunting process** moving from **initial detection** to **detailed assessment** of the attack's impact, supporting effective **incident response** and **forensic analysis**.